Tighten Codex: sensor service bench

Package `tighten_codex/`; GAR ID `tighten-codex`; engine: pinned Newton 1.5.0 with the MuJoCo-Warp solver.

Every number below is labelled by where it comes from:

- drawing/datasheet: a manufacturer document;
- reference: handbook knowledge, not measured here;
- prior: an engineering choice with no source;
- numerical: a solver-conditioning choice;
- measured: read from this simulation's own runs, with the case named.

Nothing is a specimen test of the real parts.

1. Activity and workcell

A technician robot services a compact photoelectric sensor on a bench beside a stopped conveyor. The job has three parts:

- Sensor: mount a SICK W4F-size sensor on its SICK BEF-W4-A bracket, roll it in the bracket's arc slot until its light spot lands on a white reference flag, then lock it with two prestarted M3 socket-head screws.
- Frame corner: secure the 20 mm aluminium frame corner that carries the sensor with two prestarted M4 screws: one forward into the post's T-slot, one upward into the rail's T-slot.
- Tools: each screw is finished when its preset torque driver clicks. The green Wera 7440-type driver (2.5 mm hex, 0.35 N·m) is for the M3s; the yellow HansTool 0176-type (3 mm hex, 0.60 N·m) is for the M4s.



Station overview

The bench carries:

- the post and rail with the sensor on its bracket;
- a white reference flag on a slide;
- two rubber V-groove tool holders, each beside the arm that uses it;
- a close inspection camera on a bench-clamped mast, and an overview camera on a 30 × 30 mm profile post;
- a service card and a station sign.

Behind it are the stopped guarded conveyor, service cabinets and an emergency stop. The room context is visual only. The two catalog arms stand on the bench front edge facing the bench.

The work is "tighten" in the grounded action inventory. Of 529 rows, 168 name a screwdriver, 326 a wrench and 47 a torque wrench. This environment covers preset screwdrivers on socket-head screws only; a wrench-on-bolt-and-nut variant is not built (§10).

2. The mechanisms

2.1 Sensor mount (from the SICK drawings)

The BEF-W4-A front leg has a round Ø3.4 mm hole and, 9 mm away, a 30° arc slot centred on that hole (drawing). The W4F's two M3 holes run through its 12.1 mm depth, parallel to the beam (drawing).

So the sensor sits face-flat on the plate and rolls about the pivot M3, while the second M3 rides in the slot. The model has exactly that: a revolute joint about the pivot-screw axis, with ±15° limits at the slot ends. Rolling moves the light spot sideways on a 10.1 mm-radius arc (±2.3 mm at the slot ends); it does not change where the beam points.

On a real installation this slot mainly sets the sensor's roll relative to the direction of material travel. Here the `aim` work order uses it to put the spot on a flag 3.5 mm to one side. That is geometrically faithful but an atypical service reason, and the instruction says "roll the sensor in its arc slot".

Both M3s thread into a steel clamping strip behind the plate (prior; the drawing does not show it), so their reactions act on the sensor. The sensor's 30 g (datasheet) is split between a 24 g housing and 6 g of pigtail and M8 plug hanging 50 mm below the housing centre. That lowers the centre of mass by 10 mm. A loose sensor still rests against the +15° slot end under its own weight (the construction settle confirms it, 14.88° at the start of `sensor_aim_lock`), so every scene starts there. The bracket's 30 g (2 mm stainless, about 1890 mm² of legs; drawing plus reference density) is part of the rail body.

2.2 Frame corner: a loose corner droops

The 280 mm HFS5-2020 rail (0.14 kg) plus the BEF-W4-A (0.03 kg) and the sensor is a cantilever over the corner bracket's 44.5 mm horizontal leg. It is held against tipping only by the upward M4, whose washer sits 0.5 mm below the leg when loose.

The rail can slide along its slot, lift, pitch and yaw. Its roll about its own axis is locked (reduced, disclosed). Yaw is keyed by a modelled 10 × 7.6 × 1.5 mm bracket tab in the rail's 8 mm bottom slot opening, with 0.2 mm of clearance per side.

The rail's seat on the leg is the rail joint's own one-sided stops about the leg's far top edge:

- Lift stop: the rail cannot go below the leg's top face (lift ≥ 0), but can rise freely.
- Nose-up stop: it cannot rotate nose-up into the leg face (pitch ≥ 0).
- Droop: nose-down it swings until the upward M4's washer, bearing on the leg's underside, pulls its T-nut end down.
- Slide: ±3.0 mm, where the upward M4's 4 mm shank meets the ends of the bracket's 10 mm slot (drawing).
- Yaw: ±6.15 mrad, where the key tab's far corner (32.5 mm from the pivot) takes up its 0.2 mm clearance (drawing).

This is exact rigid geometry represented as stops (reduced, disclosed). Contact is not used for these limits because Newton converts a contact's force-space stiffness using MuJoCo's direction-averaged body inverse weight: a stop carried by a light body (a 2 g screw) in front of a heavy one (the 0.2 kg rail) acts about 20× softer than set. As box contact, the key tab lets a 100 N screw pull yaw the rail 2–3× past its clearance; the slot-end stop as shank contact, set to 5e7 N/m, delivers about 0.7 MN/m, and 5e8 tunnels. The stops' damping is 1e-4 s × stiffness: at 1e-5 s the loose rail bounces off its slot end (1.55 mm spring-back) and leaves the slot. The seat's normal load, which sets the rail's sliding friction, is the rail's statics: weight plus every external vertical force, including the washer pull.

Measured (`frame_clamp`, `rest_long`):

- Droop: the loose rail rests drooped by 36.9 mrad nose-down, with 22 N on the washer. A hand estimate gives about 37 mrad and about 16 N.
- Support: a supporting hand holds it level (pitch 0.97 mrad).
- Tightening: tightening the upward M4 while supported leaves it level, and it stays level after the support is released (pitch -0.19 mrad, yaw 0.00 mrad at 679 N).

The corner bracket (15 g) slides on the post (x), toward the post face (y) and vertically. The post faces limit it by contact. The forward M4's 4 mm shank in its 4.5 mm hole limits x and z to ±0.25 mm, as joint stops for the same reason. Its rotations are locked (reduced, disclosed).

2.3 Screw joints

Each screw is a chain:

- a nut anchor, fixed to the part it threads into (the sensor's clamping strip, the rail's T-nut or the post's T-nut);
- an axial carrier on a prismatic joint;
- the screw on a revolute joint.

A MuJoCo joint equality ties advance to turn (pitch 0.5 mm M3, 0.7 mm M4; standard). The head bears on the clamped part through two small points on the non-rotating carrier, over two dedicated bearing lands. That is where the joint stiffness is set.

Friction is a MuJoCo friction-loss row on the screw coordinate, with its capacity set every tick from solved loads (the ISO 16047 form):

- Thread: $(d2/2)[\tan(\lambda + \operatorname{atan}(\mu t/\cos 30°)) - \tan \lambda]$ times the axial thread load, meaning the net axial force on the screw chain.
- Under the head: $\mu b\ Db/2$ times the solved head load.

Self-locking. The lead angle (3.4° M3, 3.6° M4) is below the friction angle (9.2°), so a prestarted screw cannot be turned by axial load. While nothing touches a screw's head, its turn and axial slide are held by stiff zero-width stops. Measured: a 100 N axial push turns an M3 at most 0.061 mrad and a 100 N pull 0.056 mrad (fault: None); without the stops a 100 N push turns it 0.27 rad.

Sticking (Coulomb). With a bit on the head, the screw is held by the same stop while the solved bit moment stays below the thread plus under-head capacity. It slips at that capacity, against the friction row, and sticks again once it has stopped (below 0.02 rad/s). MuJoCo's friction-loss row on its own creeps under steady load below its capacity: held at 94 % of the setting for 30 s, the loaded M3 turns 67.6 mrad and gains 24 N with the row alone. With the stick: held at 86 % for 1 s the screw turned 0.00 mrad; at 94 % for 30 s it still advanced 6 mrad from 365 N, when the bit-in-hex contact moment's peaks cross the breakaway (§10).

2.4 Joint stiffness

A contact pair acts as its two shapes' stiffnesses in series. The bearing-point and land stiffnesses are chosen so each stack matches the stack-up below (reference constants, drawing geometry, no interface compliance):

Stack	Elements (MN/m)	Stack-up	Model, measured
M3 (head on plastic ear)	plastic ear column 3.6–12 (3–10 GPa), steel screw 64	3.4–10.2 MN/m	4.53 / 4.53 MN/m (pivot / arc slot)
M4 upward (washer on leg, T-nut on rail lips)	screw 222, washer 12 900, leg 2 460, two slot lips in bending 76	≈ 55 MN/m	7.85 MN/m
M4 forward (head on leg, T-nut on post lips)	screw 246, leg 1 540, two slot lips 76	≈ 56 MN/m	9.38 MN/m

For both M4s the dominant element in this estimate is bending of the two thin aluminium slot lips under the T-nut. It leaves out interface (roughness) compliance, so real joints are softer than the stack-up. The stack-up is an estimate, not a calibration target.

What the model's stiffness is made of. The upward M4's clamp loop is the head/washer contact on the leg's underside in series with the rail's seat stop on the leg. The forward M4's loop is its head contact in series with the bracket's contact on the post face.

Two conversion effects in the solver make stops and contacts much softer than set (without the measures below the upward M4 reads 3.1 MN/m, with 117° of rotation from 20 N to the click):

- A head contact on the screw's 1 g axial carrier. Newton turns a force-space contact stiffness into a MuJoCo solver time constant using the bodies' inverse weights. On so light a body the time constant falls below MuJoCo's floor of 2·dt, and MuJoCo clamps it, softening the contact.
- The seat stop has the same problem. At the rail's pivot edge its lift coordinate carries only ~17.5 g.

The measures, each checked on the stiffness probe and on the loose-rest stillness gate:

- Reflected axial inertia (derived). Each screw's axial slide carries the real screw's rotational inertia reflected through its lead, $I\,(2\pi / \text{pitch})^2$: about 1.0 kg for the M4 and 0.6 kg for the M3. The thread

couples the two motions, so the head contact sees the inertia it really moves.
- Thicker hidden lands. The bearing lands are 3 mm deep: a stiffer head contact tunnels through 0.5 mm lands.
- Upward bearing 2.5e7 N/m. 3e7 and above spike or run away (4e7 to 3,900 N, 6e7 drives the loose rail unstable).
- Seat stop 5e7 N/m at 1e4 N·s/m, with a 0.15 kg armature on the lift coordinate (numerical). Without the armature, no seat stiffer than about 2e7 stays stable; 5e7 with 500 N·s/m bounces the loose rail (washer 0–68 N).
- Washer damping 13,000 N·s/m, with a 0.2 kg bracket armature (numerical). This gives a damping ratio of about 0.7, with the time constant above 2·dt. Lighter settings let the loose rail's timestep-bound chatter return in some 10 s rests (0.1 kg at 6,000 N·s/m: one 10 s rest failed at 1.19 µm).

Measured:

- Upward M4: 52° from 20 N to the click.
- Forward M4: 36°.

Measured on the case runs, the model is still well below the stack-up. Snug to click is a few tens of degrees, not a third of a turn. No specimen measurement of this joint's snug-to-click angle exists here, so that angle is not calibrated.

2.5 Clamp friction per material pair

Clamp friction is also friction-loss rows, on the sliding coordinates: rail along its slot and in yaw, bracket along and toward the post, and sensor about its pivot. Each row's capacity is µ times the solved normal load across each interface that shears in that direction. The contact faces themselves are near-frictionless in the solver: MuJoCo-Warp contact friction creeps under steady shear, and has no no-slip pass.

A friction-loss row alone also creeps below its capacity. It acts like a very stiff damper: the clamped rail moves at about 1.8 µm/s per newton. With rows alone a "clamped breakaway" reads 213 N at clamp µ 0.30, 0.45 and 0.60 alike: it measures creep, not friction.

The rail's slide and yaw and the bracket's x and z therefore stick (Coulomb):

- While stuck, a zero-width joint stop holds the coordinate, and the kinetic friction row is off, so static friction is the stop alone.
- The force the stop carries is read from the solver's own joint-limit constraint row. Checked on a stuck rail under 150 N: limit row 118.7 N plus friction row 31.3 N.
- The coordinate slips when that force reaches the friction capacity, and sticks again once it has stopped, at a stick position clipped into the joint range (otherwise a re-stick beyond the slot stop ratchets the rail out of its slot under a 50 N push).

Estimating the force instead as a calibrated stiffness times the deflection, with the friction row left on, delays the read-back by ~0.26 s through the row's viscous regularisation, so a short push above capacity is ignored and the ramp reads ~10 % high; that calibration also breaks whenever an inertia changes. Measured: the clamped rail broke away at 394 N against a computed capacity of 393 N (rail 401 N, bracket 393 N) at the 200 N/s ramp, and a 50 ms, 20 N push moved the loose rail 3.02 mm to its slot stop.

The clamped rail's yaw has a friction row of its own: (rail-on-leg µ + head-on-washer µ) × upward M4 load × 13.5 mm lever; without it a clamped rail swings to its tab stop under a sideways push. The rail rows use a near-rigid friction impedance (numerical): with the default the loose rail's yaw creeps 0.1 mrad/s.

Pair	µ	Range used	Source
Thread, steel on zinc-plated steel	0.14	—	reference, Bossard electro-zinc, low end
Upward M4 head on steel washer	0.14	—	reference, same
M3 head on glass-fibre plastic (VISTAL) housing	0.30	0.20–0.40 (§8)	reference, steel on glass-filled plastics
Forward M4 head on anodised aluminium	0.20	—	reference

Pair	μ	Range used	Source
Anodised aluminium on anodised aluminium (rail, bracket, post)	0.45	0.30–0.60 (§8)	reference
W4F plastic face on stainless plate / steel strip on plate	0.30 / 0.15	—	reference

With μb = 0.30 on plastic, the M3 torque equation gives 360 N at the 0.35 N·m click. The forward M4 gives 598 N and the upward M4 673 N at 0.60 N·m.

2.6 Preset drivers: a one-way detent clutch that clicks

Both products have a clutch that slips in the tightening direction only. The Wera 7440 datasheet gives "unlimited torque to loosen seized screws"; HansTool gives "±6 % accuracy clockwise". The model is a cam detent on the relative handle-to-bit coordinate:

- Locked: a stiff zero-width stop holds the coordinate in both directions, so there is no creep.
- Climb: when the torque through the lock reaches the setting in the tightening direction, the cam starts up its ramp. The preloaded spring pushes back with the setting torque over a 0.1 rad climb (prior). If the drive drops below the setting before the top, the cam falls back into its notch.
- Release and notch: over the top it travels on to the next notch (60°, prior) against a reduced roll-over torque (30 % of the setting, prior) and relocks there. Completing a notch is one click.
- Loosening: never releases.

A release therefore costs the setting times the climb in work (0.035 J for the M3). A clutch that released the moment the torque touched the setting would be re-released by a 0.5 mJ relock impact and left mid-notch, and the next screw would "click" at 113 N (the evaluator's load floor rejects such a click).

A driver has "released" on a screw after at least one click with its bit in that socket. Measured (`clutch_limit`, `clutch_hold`):

- Release onset: the cam started up its ramp at tick 8122 (the torque through the lock read 0.350 N·m on that first 0.21 ms tick at or above the 0.35 N·m setting), went over the top at tick 8332 and clicked with 364 N on the M3 (equation 360 N); the 2.1 ms bit moment peaked at 0.353 N·m.
- Below the setting: held at 86 % of the setting for 1 s, and at 94 % for 30 s, the clutch did not release (clutch motion measured from the tick the hand reaches the hold torque; 1 s at 0.30 N·m: 0 clicks, clutch -0.70 mrad, screw 0.00 mrad; 30 s at 0.33 N·m: 0 clicks, clutch 0.20 mrad, screw 5.73 mrad). A friction-loss clutch, by contrast, creeps 4–20 mrad/s below its setting and can register a "release" without reaching it.

2.7 Tools, holders and contact

The drivers are free bodies with product masses: 180 g Wera, and 38 g HansTool including an authored 23 mm grip collar (datasheet plus prior). Each lies in a 90° rubber V-groove holder, at its computed rest height.

The collision handles are cylinders. The Wera's hexagonal anti-roll section is visual only.

Handle bounce. One contact damping for both handles (300 N·s/m) gives restitution 0.85 for the 180 g M3 and 0.32 for the 38 g M4: mass, not material, sets the bounce. The damping is therefore set per driver toward a common target restitution of 0.5. That is a generic plastic-on-hard-surface value from a web table (web-sourced); there is no drop test of these handles.

- M3: calibrated by dropping it in the model: 1800 N·s/m gives 0.49 from 50 mm and 0.50 from 20 mm, and it is within 0.2 mm of rest after 0.35 s (at 300 N·s/m it is still bouncing after 0.8 s).
- M4: it lands on its collar and handle sections in turn, so its first bounce is set by geometry. It measured 0.82, 0.14, 0.23, 0.32, 0.36, 0.10 and 0.23 at 100, 150, 200, 300, 600, 830 and 1200 N·s/m, with no trend. It therefore gets the M3's damping ratio (damping ∝ √mass), 830 N·s/m: first bounce 0.10, settled after 0.38 s.

3. Tasks and evaluation

The six recipes in `task.py` share the same products, screws and robots. Each is one work order, in plain words, to the policy:

Task	Start state	What must be true at the end
`sensor`	Frame secured; sensor loose at its slot end; flag in place	Both M3s clicked and retained; flag back; beam detects it
`aim`	Frame secured; sensor loose at the slot end, spot off the flag (flag −3.5 mm)	Sensor rolled until the spot is on the flag, both M3s clicked and retained, detection after release
`frame`	Corner loose, rail drooped onto the upward M4	Both M4s clicked and retained; rail at its slot reference (±1 mm), level and square (±3 mrad)
`support`	As `frame`, rail also 1.5 mm along its slot	As `frame`, rail within ±0.5 mm
`mixed`	Everything loose, rail drooped; flag home stop	All four screws clicked by their own drivers and holding hands-off; rail level; reference flag at its home and detected there; nothing touching the assembly (rules below)
`access`	As `frame`, plus a service guard around the upward M4	Upward M4 clicked and retained through the opening

The evaluator (`TightenEnv._evaluate`) checks mechanism-level signatures only:

- Driver released on the screw: the preset driver's clutch completed at least one click while its bit was in that screw's socket.
- Load retained: after the bit leaves the socket, the solved head load stays at or above 80 % of the load the equation gives at the driver's setting. It must hold for 0.25 s with the helical advance above 0.40 mm (M3) or 0.45 mm (M4), and with under 0.001 rad angle drift. A recontact, load dip or drift resets the timer.
- Sensor torque: the solved M3 bit moment, averaged over 2.1 ms, never exceeds the W4F's 0.4 N·m maximum.
- Rail (frame tasks): at its slot reference, and level and square within 3 mrad of pitch and yaw. That tolerance is about 1/12 of a loose corner's measured droop.
- Optical (sensor tasks): a geometric ray from the W4F lens hits the flag.

These thresholds are authored targets, not manufacturer clamp-load specifications. Privileged forces, angles and optical truth stay out of restricted policy observations.

In the sensor tasks the work order says the frame is already secured, so the scene tightens it to its 0.60 N·m click during construction.

`mixed`. `mixed` alone uses these rules (`TightenEnv.MIXED_SCOPE`), which avoid hidden gates and an all-or-nothing score:

- Click: counted only from the screw's own preset driver, and void once its hands-off head load has read below the 80 % floor for 10 consecutive readings, or once the screw turns back more than 0.01 rad past its angle at that click, with the bit in or not. A seated screw cannot back off by itself in this model; the recorded after-click back-rotation is at most 1.76 mrad. A re-seated screw must click again.
- Retention: 10 ticks (2.1 ms) instead of 0.25 s, with no bit in the socket and no robot link, or driver a robot link touches, on what the screw clamps (sensor group for the M3s; rail, sensor, bracket and screw chains for the M4s). The verdict depends only on the last 2.1 ms after the last withdrawal.
- Rail: level within 3 mrad of pitch (`rail_in_reference` means level here). No separate slot-position or yaw gate (the key tab already bounds yaw at ±6.15 mrad); both move the beam, so they count through detection at the flag's home.
- Reference flag: back at its home, the carriage within 0.5 mm of its start (the stop holds it at −0.1 mm), and detected there, so moving the flag to meet a displaced beam does not pass.
- Detection: the spot centre on the 10 × 44 mm white face, i.e. most of the nominal 4.2 mm spot on it, instead of the whole spot. A background-suppression sensor switches when the near-range return outweighs the far one; a spot 95-99.9 % on the white flag failed the whole-spot rule. No partial-coverage switching curve is sourced, so this is a documented physical argument. The other tasks keep the whole-spot rule.

- Hands-off: nothing robot-held touches the assembly at evaluation ("hold on its own after you let go").
- Score: 1.0 only with success; otherwise 0.95 × the mean of four screw credits. A screw's seat is 0 for an M3 whose 2.1 ms bit moment exceeded 0.4 N·m, 1 for a clicked screw whose last hands-off load is at or above the floor with advance, else 0.9 × last hands-off load / setting load. The upward M4's seat × rail-level grade from the last hands-off pitch (0 at the 36.9 mrad free droop, 1 within 3 mrad); the forward M4's seat alone (it clamps the non-rotating bracket to the post and does not set the rail's attitude). M3 seat × (0.75 + 0.25 × detection grade). Detection grade = aim × return: the aim is judged against a flag at its home position (1 when the spot centre would be on it, else twice the spot's share there), so moving the flag never raises it; the return share is 1 within the 0.5 mm band and falls linearly to 0 at 10 mm (the detected LED is lit to about 7 mm), so a flag left short of home is graded rather than scored as never returned. Head loads are the last hands-off reading, lowered but never raised by readings under a bit or hand, so loosening under a tool shows and pushing cannot inflate; pitch is read only hands-off. A driver left resting in a socket still blocks that screw's hands-off reading (the screw then scores from its last hands-off load). No term for order, motion, contact, support, tool pickup or flag retraction.
- Scene cues: a black-oxide steel stop on the worktop at the flag track's return end (flag rests 0.1 mm past home, inside the detection window; the slide's joint, limit, damping and friction are unchanged), and, render only, the W4F's yellow output LED lit from the geometric ray and a yellow ring on the M4 driver's collar.

The other five recipes keep the rules above unchanged.

4. Robots

All robots come from the shared catalog (`robots/`). Arm and hand URDFs, gains and the environment's own `tighten` controller profile are in `robots/models/{fr3,panda}/robot.json`. The world owns the mounts (`tighten_codex/kinematics.py`). The IK is the shared `SerialKinematics`, built from each instance's own URDF.

`--robot`	**Rig**	**Arms**	**Tool cradles**
`dual-fr3` (default)	`robots/rigs/tighten-fr3.json`	two FR3 + Franka Hand	M3 right (x 0.36), M4 left (x −0.36)
`single-fr3`	`robots/rigs/tighten-fr3-single.json`	right FR3 + Franka Hand	both right (M3 x 0.36, M4 x 0.50)
`dual-panda`	`robots/rigs/tighten-panda.json`	two Panda + Franka Hand	as dual FR3
any validated setup	a `gar.robot-setup.v1` object	left and/or right catalog arm (FR3 or Panda) on its bench mount, Franka Hand each	by the arms present

- Mounts face the bench. Both arms stand on the bench front edge at x = ±0.43 m, yawed +90°, so the shared start pose puts the hands 0.86 m apart. With yaw-0 mounts the left hand would start beside the right elbow, and a two-arm `CartesianPose` would stop on the arm-arm guard.
- Tools beside their work. The green M3 driver lies on the right, next to the sensor at x = 0.27 m. The yellow M4 driver lies on the left, by the corner at x = 0. A single arm gets both cradles on its own side, so `single-fr3` can fetch either tool. Measured: the single right FR3 lifts the green driver from its own holder by 70.9 mm through public commands (§9).
- Open selection. Any `gar.robot-setup.v1` whose arms are named `left`/`right`, use `left_arm_mount`/`right_arm_mount`, are catalog arms with a bench tool-centre point (FR3 or Panda), and carry a `franka-hand@1`, is accepted (`validate_setup`). Tested with a single-Panda setup; others are rejected with the reason.
- No humanoid. The catalog G1 is rejected with a measured reason. From its URDF, standing at the bench front edge its shoulder is about 1.07 m high and its straight-arm reach to the palm about 0.43 m. The driver grips in the cradles lie about 0.74 m away, and the upward M4 is driven from below at about 0.58 m. The repository has no balance or whole-body controller qualified for manipulation (whiteboard's G1 stands on joint servos only). A free-standing G1 would fall. A G1 pinned at the pelvis would be a fixture shaped like a humanoid, which misrepresents what a humanoid can do at this bench. The humanoid is therefore out of scope for this environment by that measured reason, not left unbuilt by oversight. A humanoid technician needs a qualified balance controller and a bench layout inside its reach.

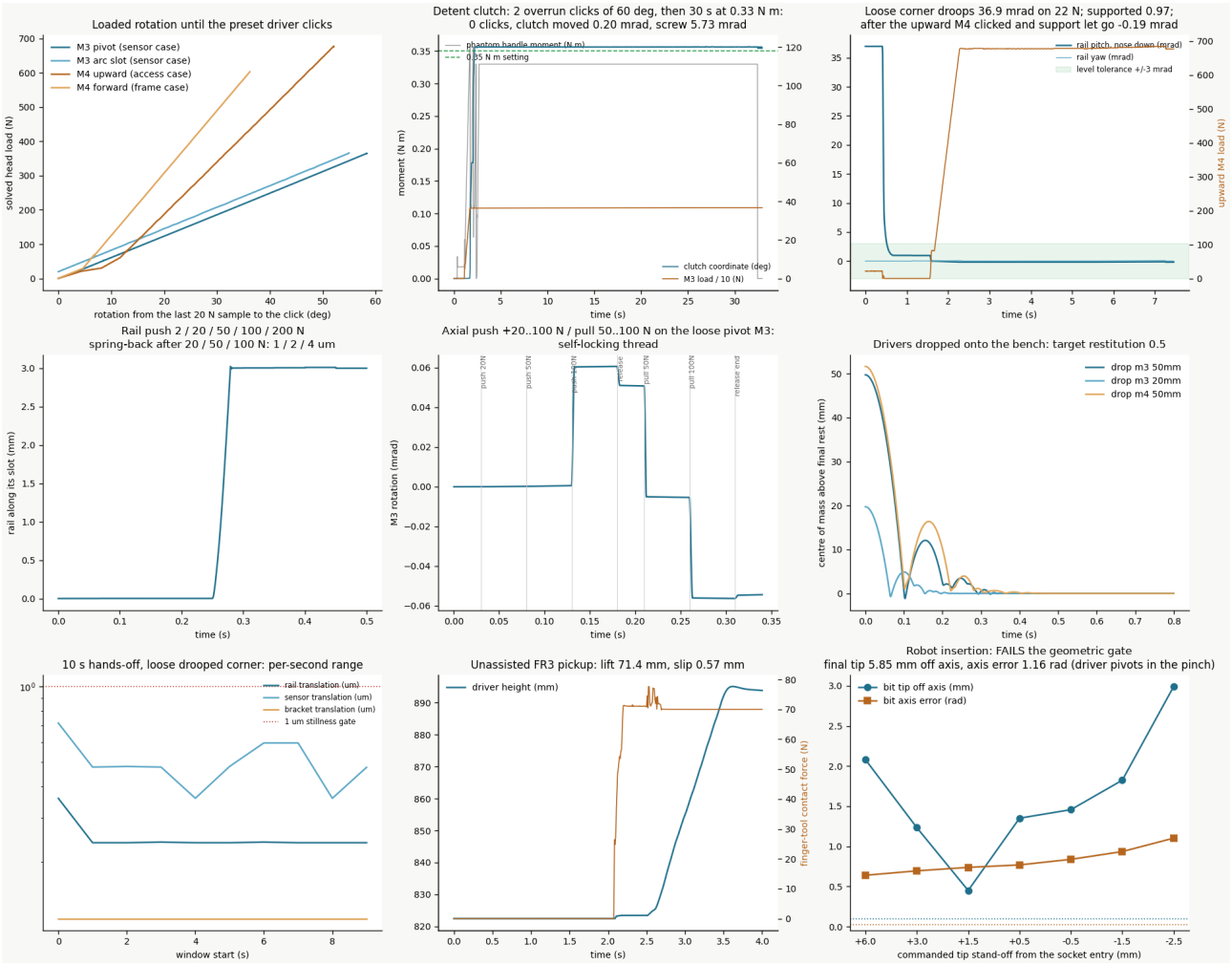
5. Assisted component evidence

Each case is one continuous rollout from the construction settle. Its films replay the saved 25 ms poses without any physics step. Every film is labelled "assisted component test" and names its assistance.

The test guide centres the M3 bit on the socket's settled position (the pre-tightened frame moves it 0.1–0.2 mm, enough to leave a bit aimed at the drawing position on the socket rim, slipping). The guide withdraws at 0.1 m/s (prior: a careful hand, 20 mm in the 0.2 s withdraw phase): a stepped withdrawal pulls the bit off the upward M4's head at about 1.2 m/s, breaks the stick of the still-loose bracket (a 148 N row force against its 1.9 N capacity) and slides the bracket, with the rail clamped to it, 0.29 mm. With the slow withdrawal no withdrawal moved the bracket more than 7.6 µm or yawed the rail more than 0.11 mrad. The clutch-hold case uses a speed-limited hand, not a torque source.

Case	What it shows	Measured
`sensor_aim_lock`	Pre-tightened frame. The loose W4F rests at its slot end; a phantom hand rolls it until the spot is on the flag and holds it; the flag slides aside; the guided green driver drives both M3s to the click; release; flag back; loose and clamped breakaway	aimed -0.239 rad, spot error 1.49 mm; M3 365 / 366 N (floor 288 N); 2.1 ms bit moment ≤ 0.357 N·m; detected after release, spot error 1.07 mm; breakaway loose 0.0044 N·m, clamped 1.93 N·m
`frame_clamp` (`support` order)	The drooped loose corner; loose breakaway; phantom support lifts it level and back along the slot; the upward M4 to the click while supported; support released; front-loose breakaway; the forward M4 to the click; clamped breakaway	droop 36.9 mrad on 22 N; supported 0.97 mrad; after release -0.19 mrad pitch, 0.00 mrad yaw; M4 677 / 603 N (floors 538 / 479); breakaway loose 1.26 N, front loose 1.51 N, clamped 394 N
`access`	The upward M4 through the service guard, rail supported	676 N; guard contact ticks 0
`clutch_limit`	Pivot M3 to the click, 0.5 s overrun (clicks), eased into its detent, 1 s at 0.30 N·m	the cam started up its ramp at tick 8122 (the torque through the lock read 0.350 N·m on that first 0.21 ms tick at or above the 0.35 N·m setting), went over the top at tick 8332 and clicked with 364 N on the M3 (equation 360 N); the 2.1 ms bit moment peaked at 0.353 N·m; 2 clicks; hold: 0 clicks, clutch -0.70 mrad, screw 0.00 mrad
`clutch_hold`	As `clutch_limit`, then 30 s at 0.33 N·m (94 %)	0 clicks, 0 releases, clutch 0.20 mrad, screw 5.7 mrad (no film)
`rest_long`	10 s hands-off, `mixed` scene: loose drooped corner, loose sensor, both drivers in their holders	largest per-second range in the last second: rail 0.24 µm, sensor 0.48 µm, bracket 0.12 µm (gate 1 µm)

Identical-code repeats (§8) give run-to-run spread. The measured figure is below; the case summaries hold every-tick maxima.



Measured results

6. Robot component evidence

The FR3 demonstrations use catalog joint targets from the shared IK and physical finger contact only.

- Pickup, unassisted (passes). The right FR3 opens over the green driver in its holder, descends, closes and lifts it: lift 71.4 mm, slip in the hand 0.57 mm over 0.5 s, finger force 70 N.
- Bit insertion (fails its gate). The FR3 lifts the driver and turns it bit-forward (via yaw π; the other way puts joint 7 on its limit). Then, with the flag slid aside (phantom) and aligned from the simulator's measured tool pose (privileged), it steps the bit toward the pivot M3's socket.
  - Before the approach: the tip is 1.9 mm off axis, with the driver tilted 0.17 rad in the fingers.
  - During the approach the tilt grows to 1.16 rad, as the driver pivots in the two-finger pinch.
  - At the end the tip is 5.9 mm off the axis and -10.0 mm from the entry, with 0 socket contact(s).
  - The geometric gate (inside the socket, on axis within 0.1 mm, coaxial within 1.5°) fails.

No robot has seated a bit or tightened a screw.

7. Dynamics probes

Probes push the loose rail, the screws and the sensor into their stops and drop the drivers on the bare worktop, against the final scene:

Probe	Result
`rail_release`: loose rail pushed 20 / 50 / 100 N into its slot stop, then released	1 µm / 2 µm / 4 µm spring-back; fault None
`rail_push`: 2 N, release, then 20…200 N into the stop, release	at 200 N 3.009 mm along the slot (stop at 3.000 mm); after release 2.998 mm. The loose 15 g bracket carries the ~2 mJ of elastic energy across its own 0.5 mm hole clearance. Fault None
`screw_push`: ±20…100 N axial on the unseated pivot M3	a 100 N push turns it at most 0.061 mrad and a 100 N pull 0.056 mrad (fault: None)
`sensor_push`: loose sensor to the far slot end, 0.5 N·m into the stop, release	fault None
`rest_task support` / `aim`: 1 s hands-off	fault None / None
`drop_m3_50mm`: driver released above bare worktop	e = 0.49 (12.1 mm rebound of 49.7 mm), within 0.2 mm of rest after 0.32 s; max energy gain -3.4e-07 J
`drop_m3_20mm`: driver released above bare worktop	e = 0.50 (4.9 mm rebound of 19.7 mm), within 0.2 mm of rest after 0.21 s; max energy gain -3.4e-07 J
`drop_m4_50mm`: driver released above bare worktop	e = 0.11 (0.6 mm rebound of 51.7 mm), within 0.2 mm of rest after 0.40 s; max energy gain -7.8e-08 J

8. Sensitivity

End-to-end reruns of the assisted cases with one reference coefficient changed across its stated range (Section 2.5), and identical-code repeats. The equation load follows the head coefficient; the clamped breakaway follows the clamp coefficient.

Variant	Case	Result	Checks
baseline M3 head μ 0.30	`clutch_limit`	click at 364 N (floor 288 N, equation 359 N)	all pass
identical-code repeat	`clutch_limit`	click at 364 N (floor 288 N, equation 359 N)	all pass
M3 head μ 0.20	`clutch_limit`	click at 474 N (floor 374 N, equation 467 N)	all pass
M3 head μ 0.40	`clutch_limit`	click at 296 N (floor 234 N, equation 292 N)	all pass
baseline clamp μ 0.45	`frame_clamp`	M4 677 / 603 N; clamped breakaway 394 N; after release pitch -0.19 mrad	all pass
identical-code repeat1	`frame_clamp`	M4 677 / 603 N; clamped breakaway 394 N; after release pitch -0.19 mrad	all pass
identical-code repeat2	`frame_clamp`	M4 677 / 603 N; clamped breakaway 394 N; after release pitch -0.19 mrad	all pass
clamp μ 0.30	`frame_clamp`	M4 679 / 604 N; clamped breakaway 302 N; after release pitch -0.19 mrad	all pass
clamp μ 0.60	`frame_clamp`	M4 677 / 603 N; clamped breakaway 485 N; after release pitch -0.19 mrad	all pass

`frame_clamp`'s evaluator runs at the very end, after the clamped-breakaway probe has slid the rail 0.24–0.37 mm (more at higher μ, because the ramp ends further above the static load). Before that probe the rail is 0.12–0.13 mm from reference in all five runs. After it, the 0.5 mm `support` gate is met at 0.37–0.49 mm; the μ 0.60 variant passes by 6 μm. The gate was not loosened.

9. Public interface, speed, history and checkpoints

Timing. A timing run drives each named setup only through the public policy path, with the default two 960 × 540 RGB cameras and every-tick history. It runs a sweep of free-space commands of 0.01–2 s, then a guarded `CartesianPose` per arm from reset, then the right-arm grasp of the green driver by IK waypoints sent as joint targets. The host was shared with other work, at a 1-minute load average of 5–32 on a 16-core quota, and the GPU was held at 99–100 % by another job's renders.

Setup	Construct + first observation	Wall ÷ simulated, 0.01 / 0.05 / 0.1 / 0.2 / 0.5 / 1 / 2 s commands	Shortest command ≤ 200×	Cartesian from reset	Right-arm grasp: lift, max wall ÷ sim of 0.5 s commands
dual FR3	147.3 s	1797 / 378 / 178 / 204 / 196 / 194 / 124	0.5 s	left: reached (0.13 mm); right: reached (0.13 mm)	71.0 mm, 146× (passed)
single FR3	90.6 s	1809 / 358 / 179 / 159 / 166 / 155 / 148	0.1 s	right: reached (0.13 mm)	70.9 mm, 79× (passed)
dual Panda	57.7 s	2884 / 379 / 182 / 168 / 179 / 157 / 143	0.1 s	left: reached (0.03 mm); right: reached (0.03 mm)	70.1 mm, 146× (passed)

Within the same dual-FR3 run, a physics tick took 40 ms during the 0.2–1 s commands and 15–20 ms later. Two further dual-FR3 runs on the same host, back to back, through the same public path and commands:

Wall time	Run 1 / run 2
Construction and first observation	47.1 s / 44.6 s
0.01 s command	18.7 s (1873×) / 13.2 s (1321×)
0.05 s command	16.7 s (334×) / 13.4 s (268×)
0.1 s command	13.7 s (137×) / 13.2 s (132×)
0.2 s command	16.3 s (82×) / 14.1 s (70×)

Where the time goes, measured on this host.

- One two-camera observation, about 15–18 s:
  - starting the render worker until ready: 8.7–9.1 s (Kit loading extensions);
  - the first camera's first render: 3.5–5.7 s (materials and acceleration structures);
  - the second camera: 0.1–0.2 s;
  - closing the worker: 2.3–3.0 s.
- The every-tick history: with a full commit per tick a traced physics tick takes 25.1 ms against 11.7 ms untraced, 9.7 ms of it an fsync per tick in the trace store.

How the time is kept down. No engine stays resident, and no observation or sample is dropped:

- The worker starts while the action's physics runs. `newton_tighten_codex.py` starts this operation's worker when the action begins, and warms it with one discarded render of the pre-action pose. The observation then uses it and releases it. A failed action closes it, as does an unconsumed one after 60 s. There is never more than one.
- The worker is released right after capture, in a reaper thread, so the reply does not wait for Kit's shutdown. The backend's close joins every reaper.
- Trace commits. `PhysicsTraceStore` has an opt-in `synchronous="NORMAL"`, which tighten uses. The default stays FULL, so other environments are unchanged. Every tick is still committed; WAL with NORMAL survives an application crash, and only an OS crash can roll back the latest commits. A traced tick takes 15.8 ms.
- Self-collision filtering. The FR3 self-collision pairs in Franka's own SRDF are filtered (`tighten_codex/robot_collision/`): they are 720 of the 838 resting contact candidates, and 204 remain.

What remains. For commands of 0.05 s or less, the ~9 s start and ~4 s warm-up cannot hide behind physics. Those commands stay above 200× unless an engine stays resident between operations, which the cleanup requirement rules out.

Checkpoints. The backend saves the full simulator state and evaluator tracking to the session directory (`save_checkpoint`, `restore_checkpoint`, `list_checkpoints`; labels `[A-Za-z0-9_.-]{1,64}`). The server answers unsupported backend capabilities with 501 and a JSON error instead of an empty reply. Checked on a live served session over HTTP: save, a 0.1 s arm move, list, and restore returned the tick and every joint exactly (largest joint difference 0.0); a malformed label got a 400 JSON error.

Live managed session. A dashboard check passed all 12 dashboard and API checks on an `aim` session: exact action ticks, pause, idempotency, restricted-key denials, no browser errors.

Tests. Unit and contract tests cover:

- task resolution and config validation;
- the catalog's setups, custom-setup validation and routing, and the G1 rejection;
- evaluator success and failure paths: click, retention, level rail and the M3 limit;
- the clutch's climb, fallback and click;
- the checkpoint file round trip and the HTTP 501 reply;
- with `GAR_TIGHTEN_PHYSICS_TESTS=1`, a resting scene and a public action on every named setup.

Result: 17 passed, including the two GPU tests.

Installed wheel. The installed wheel inventories, inspects, discovers all robot setups, resolves the tasks and constructs every setup from outside the checkout.

Running it. `gar inspect tighten-codex` and `gar tasks inspect tighten-codex` list the robot choices and the six tasks; `gar run tighten-codex --task mixed` starts a session (`--robot single-fr3` or `dual-panda` for the other setups).

10. Limitations

These are the gaps between this simulation and the real bench. Each is measured, and none is hidden behind a gate.

- M4 joints are softer than the stack-up. They measure 7.9 MN/m upward and 9.4 MN/m forward, against an estimated ~55 MN/m that leaves out interface compliance. The rotation from snug to click is 52° upward and 36° forward (§2.4). Two conditioning armatures make this possible: 0.15 kg on the rail's lift coordinate and 0.2 kg on the bracket. They are virtual inertia on those coordinates only, and add no weight.
- Screw creep under a steady hand near the setting. Held at 94 % of the setting for 30 s, the M3 advanced 6 mrad and gained 2 N. It advances only when the bit-in-hex contact moment's peaks cross its breakaway; the screw's reflected axial inertia (§2.4) keeps that small. At 86 % it holds. The clutch does not creep or release in either hold.
- The clutch lock rings. When the hand lets go of the handle, it rings on the clutch's lock spring by about ±10 mrad at about 20 Hz before settling. The lock is a stiff spring behind a light handle, lightly damped. Creep below the setting is therefore measured from the tick the hand reaches its hold torque. Measured from the unloaded rest, the ringing sampled 0.2–12 mrad ("clutch_change_from_unloaded_rest_rad" in the summaries). No click or release occurs in it.
- A stuck clamp's own compliance. A stuck clamp coordinate yields a few µm to tens of µm under load, at its stop's MuJoCo static stiffness. Slip is decided on the solver's force read-back (§2.5).
- Robot work is not solved. No robot tightens a screw. The scripted FR3 insertion fails its own geometric gate (§6): in a two-finger pinch the 180 g driver pivots about the pinch axis. The robot pickup is unassisted and passes.
- Assistance in the component proofs. These are all labelled in every film and summary:
  - a test-only guided driver, centred on the settled socket by test stages;
  - a speed-limited phantom hand on the handle;
  - phantom support for the sensor and the rail;
  - a phantom flag slide.
- Reduced mechanics (joints standing in for contact, stated where used):
  - the rail's seat, nose-up, slide and yaw stops;
  - the bracket's hole stops, with its rotations locked;
  - the rail's roll about its own axis, locked;
  - the sensor on a pivot joint with a clamp friction row;
  - the screw chain's helical equality;
  - the clutch's climb and notch law.
- No humanoid (§4). No wrench-on-nut variant: this environment covers preset screwdrivers on socket-head screws only, which is 168 of the 529 grounded "tighten" rows.
- Speed. Each observation still costs about 13–19 s of render-worker start, warm-up and capture, so commands of 0.05 s or less exceed 200× (0.05 s at 268–334× and 0.01 s at 1,321–1,873× in the back-to-back runs, §9). Keeping an engine resident would remove most of it, and the cleanup requirement rules that out. Physics alone slows 2–3× when other work holds the GPU.
- Calibration. No part here is a specimen test. Friction coefficients and the handle restitution target are reference or web values. The clutch's climb angle, notch and roll-over are priors.